Contrastive evidence, superposition, and our results: a compatibility check

Setup. Under our empirical finding $n = 2$, the response has the form

$$R(v) = F(v^\top A v), \qquad A = \sum_{i=1}^{M} w_i \, d_i d_i^\top, \quad w_i \geq 0,$$

with feature directions $d_i \in \mathbb{R}^D$, $\|d_i\| = 1$, and (under superposition) $M > D$. Sensitivity of a direction u is $u^\top A u$.

Contrastive claim. By isolating concepts via contrastive pairs and observing they steer the model more than random directions do, we have

$$d_j^\top A d_j \; > \; \mu \quad \text{for every feature } j, \qquad \text{where } \mu = \tfrac{1}{D}\operatorname{tr}(A)$$

is the average sensitivity of a uniformly random unit direction (i.e. $\mathbb{E}_r[r^\top A r] = \operatorname{tr}(A)/D$).

Question. Is this compatible with our results and with superposition ($M > D$)?

Summing the inequality. Multiply both sides by $w_j \geq 0$ (preserves the direction of the inequality) and sum over j:

$$\sum_{j=1}^{M} w_j \, d_j^\top A d_j \; > \; \mu \sum_{j=1}^{M} w_j.$$

Left side. Using $d_j^\top A d_j = \operatorname{tr}(A \, d_j d_j^\top)$ and linearity of trace,

$$\sum_j w_j \, d_j^\top A d_j = \operatorname{tr}\Big(A \sum_j w_j d_j d_j^\top\Big) = \operatorname{tr}(A \cdot A) = \operatorname{tr}(A^2).$$

Right side. Since $\|d_j\| = 1$ implies $\operatorname{tr}(d_j d_j^\top) = 1$,

$$\sum_j w_j = \sum_j w_j \operatorname{tr}(d_j d_j^\top) = \operatorname{tr}\Big(\sum_j w_j d_j d_j^\top\Big) = \operatorname{tr}(A).$$

Substituting $\mu = \operatorname{tr}(A)/D$:

$$\boxed{\operatorname{tr}(A^2) \; > \; \frac{\operatorname{tr}(A)^2}{D}.}$$

What this says about A. By Cauchy–Schwarz on the eigenvalues of A,

$$\operatorname{tr}(A^2) = \sum_k \lambda_k^2 \; \geq \; \frac{(\sum_k \lambda_k)^2}{D} = \frac{\operatorname{tr}(A)^2}{D},$$

with equality iff all eigenvalues are equal, i.e. $A \propto I$ (a tight frame). Our strict inequality just says *A is not a scaled identity* — its eigenvalues are not all equal. This is generic and easily satisfied under $M > D$. **No contradiction.**

But: a non-feature direction must beat every feature. Once A has unequal eigenvalues, let u^* be its top eigenvector. By Rayleigh,

$$u^{*\top} A u^* = \lambda_{\max}(A) \;\geq\; d_j^\top A d_j \quad \text{for every } j.$$

Under $M > D$ with generic features, u^* does not coincide with any d_j (that would be a measure-zero alignment among M directions in D-space). So there exists a *non-feature* direction strictly more sensitive than every feature.

Interpretation. The contrastive evidence "every feature beats a typical direction" is fully consistent with our $n = 2$ results and with superposition. What it *does not* rule out is the existence of a non-feature direction that beats every feature — in fact our results *force* such a direction to exist (generically). A contradiction with the strong reading of LRH only arises if one separately asserts that no non-feature direction can exceed the sensitivity of every feature. The contrastive argument alone doesn't establish that: it compares features to typical directions, not to optimized ones.

Empirical closure. The missing piece is direct: if one maximizes sensitivity over unit directions (MELBO-style) and the optimum consistently lands on recognizable feature directions, the strong reading holds. If the optimum lands on non-feature compositional directions, the strong reading fails — which is the empirical form of the corollary, without any universal-quantifier assumption.